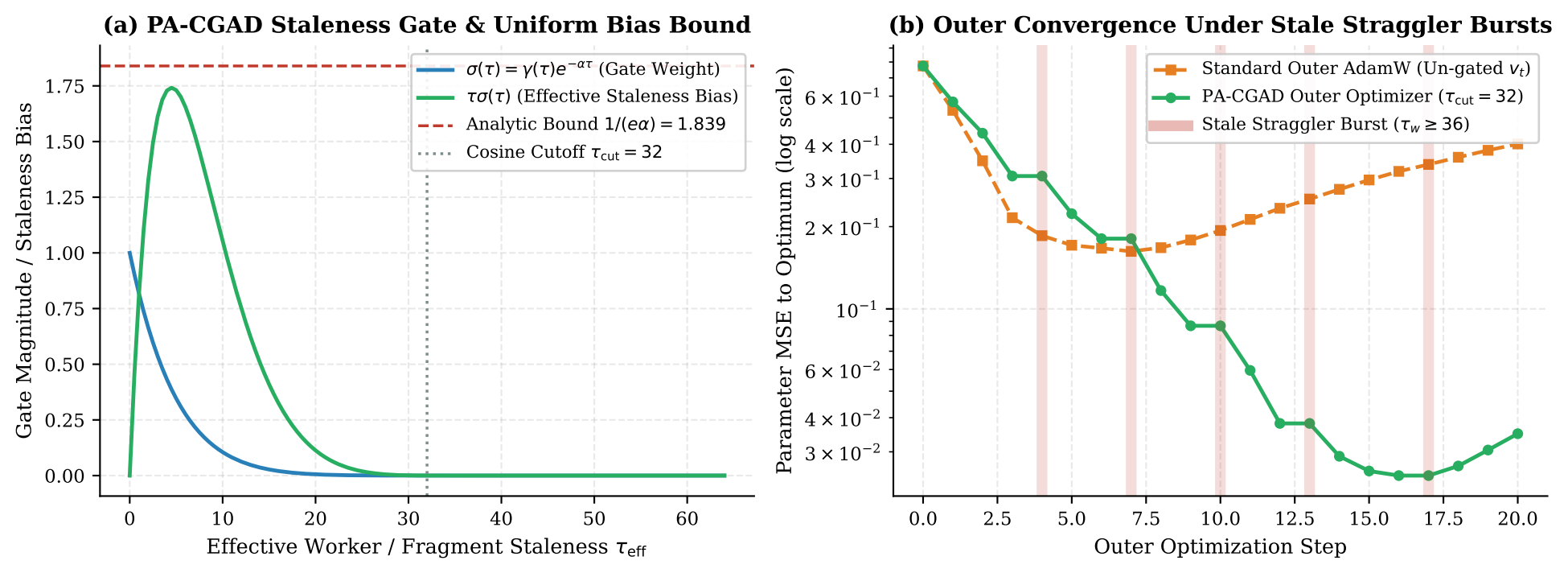
(a) PA-CGAD Staleness Gate & Uniform Bias Bound
Gate Magnitude / Staleness Bias
Effective Worker / Fragment Staleness τ_{eff}
$\sigma(\tau) = \gamma(\tau)e^{-\alpha\tau}$ (Gate Weight)
$\tau\sigma(\tau)$ (Effective Staleness Bias)
Analytic Bound $1/(e\alpha) = 1.839$
Cosine Cutoff $\tau_{\text{cut}} = 32$
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
0
10
20
30
40
50
60
(b) Outer Convergence Under Stale Straggler Bursts
Parameter MSE to Optimum (log scale)
Outer Optimization Step
Standard Outer AdamW (Un-gated v_t)
PA-CGAD Outer Optimizer ($\tau_{\text{cut}} = 32$)
Stale Straggler Burst ($\tau_w \geq 36$)
6×10^{-1}
4×10^{-1}
3×10^{-1}
2×10^{-1}
10^{-1}
6×10^{-2}
4×10^{-2}
3×10^{-2}
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0